

Metagenome Analysis 실습



2026. 농촌진흥청 슈퍼컴퓨터 기반 빅데이터 분석 교육

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발표일 : 2026년 7월 30일

목차

본 강의는 Metagenome 의 이론적 배경부터 연구설계, NGS 데이터 생산, QIIME2 파이프라인 실습 및 유의한 결과 도출을 통한 통계 분석까지 Metagenome 전 과정을 이해를 목적으로 둔다.

1. Metagenome 연구 개요
2. Metagenome 과 NGS
3. Metagenome 연구 방법론
4. 미생물 군집 다양성 분석 방법과 사례
5. **QIIME2 실습: 결과 해석 및 유의한 결과 도출을 통한 통계분석**

QIIME2 환경 세팅

<Docker 다운로드 및 Base Image 세팅>

: Docker 을 통한 환경 구축

```
# QIIME2 2026.04 image 세팅 (관리자 접속)
```

```
sudo su
```

```
docker pull quay.io/qiime2/qiime2:2026.4
```

```
sudo usermod -aG docker ubuntu
```

QIIME2 환경 세팅

<QIIME2 실행>

```
# 도커 업데이트 및 QIIME2 테스트 명령어  
newgrp docker
```

```
docker run --rm \  
--user $(id -u):$(id -g) \  
-e HOME=/tmp \  
-e MPLCONFIGDIR=/tmp/matplotlib \  
-v $(pwd):/data \  
-w /data \  
-ti quay.io/qiime2/qiime2:2026.4 \  
qiime info
```

QIIME2 환경 세팅

< 사용자 단축명령어 세팅 >

1. ~/.bashrc 파일 열고 아래 코드 작성

```
# QIIME2 Docker Runner (.bashrc)
qiime-docker() {
  docker run --rm \
  --user $(id -u):$(id -g) \
  -e HOME=/tmp \
  -e MPLCONFIGDIR=/tmp/matplotlib \
  -v "$PWD":/data \
  -w /data \
  -ti quay.io/qiime2/qiime2:2026.4 \
  qiime "$@"
}
```

QIIME2 환경 세팅

<QIIME2 실행>

- **docker run --rm** : Docker 실행, 완료시 임시 컨테이너 삭제
- **--user \$(id -u):\$(id -g)** : 학생 계정 권한으로 실행 및 저장 (이후 활용에 중요함)
- **-e HOME=/tmp -e MPLCONFIGDIR=/tmp/matplotlib** : 학생 유저 권한 시 발생하는 권한 이슈 해결
- **-v \$(pwd):/data** : 현재 디렉토리를 docker 내부 폴더와 연결
- **-ti** : 가상 터미널 및 대화형 모드 활성화
- **quay.io/qiime2/qiime2:2026.4** : 실행할 Docker 이미지 (QIIME2 2026.4 docker)
- **qiime info** : docker 가상환경내 실행할 명령어

QIIME2 환경 세팅

< 사용자 단축명령어 세팅 >

2. `~/bashrc` 적용 후 확인

```
source ~/.bashrc  
qiime-docker info
```

QIIME2 실습 – 기본 명령어 익히기

실습데이터: 인분 퇴비화 과정 미생물 군집 전이 및 안전성 검증



ISME Communications, 2025, 5(1), ycaf089

<https://doi.org/10.1093/ismeco/ycaf089>

Advance access publication: 29 May 2025

Brief Communication

Upcycling human excrement: the gut microbiome to soil microbiome axis

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본 실습 데이터의 경우 전체 데이터 셋 중 실습용 **경량화된**
서브셋

<실습데이터>

인분 퇴비화 과정 미생물 군집 전이 및 안전성 검증

인분퇴비화(HEC) 과정에서 초기 장내 마이크로바이옴이 토양 및 전통 퇴비 미생물 군집으로 전이(Succession) 되는 경로 및 안전성 검증

- Experiment Designing: 15개 퇴비 변기 혼합물을 19L 버킷에 15개에 나누어 1년간 중온 조건에서 퇴비화
- Sampling: 매주 샘플링 되었음
- Platform: Illumina MiSeq (Paired-End sequencing)
- Amplicon sequencing: 16S rRNA V4 region
- Primer: F515-R806 (Earth Microbiome Project 기반)

⇒ 본 실습은, 'SampleType' 에 따른 변화를 확인하고자 합니다.

- **Human Excrement (HE)**: 인분 원본 샘플 (초기 장내 마이크로바이옴 상태)
- **Human Excrement Compost (HEC)**: 1년간 퇴비화 진행 중인 인분 퇴비 샘플 (시간에 따른 전환 상태)
- **Food and Landscape Waste Compost (FLWC)**: 음식물 및 정원 폐기물로 만든 전통적 대조군 퇴비
- **Soil**: 자연 토양 샘플 (최종 지향점)

QIIME2 실습

< 데이터 다운로드 >

- sample-metadata.tsv | [download](#)
- demux.qza | [download](#)

프로젝트 메타정보 다운로드

```
wget -O 'sample-metadata.tsv' \  
'https://gut-to-soil-tutorial.readthedocs.io/en/2026.4/data/gut-to-soil/sample-metadata.tsv'
```

프로젝트 시퀀싱 데이터 다운로드

```
wget -O 'demux.qza' \  
'https://gut-to-soil-tutorial.readthedocs.io/en/2026.4/data/gut-to-soil/demux.qza'
```

QIIME2 실습

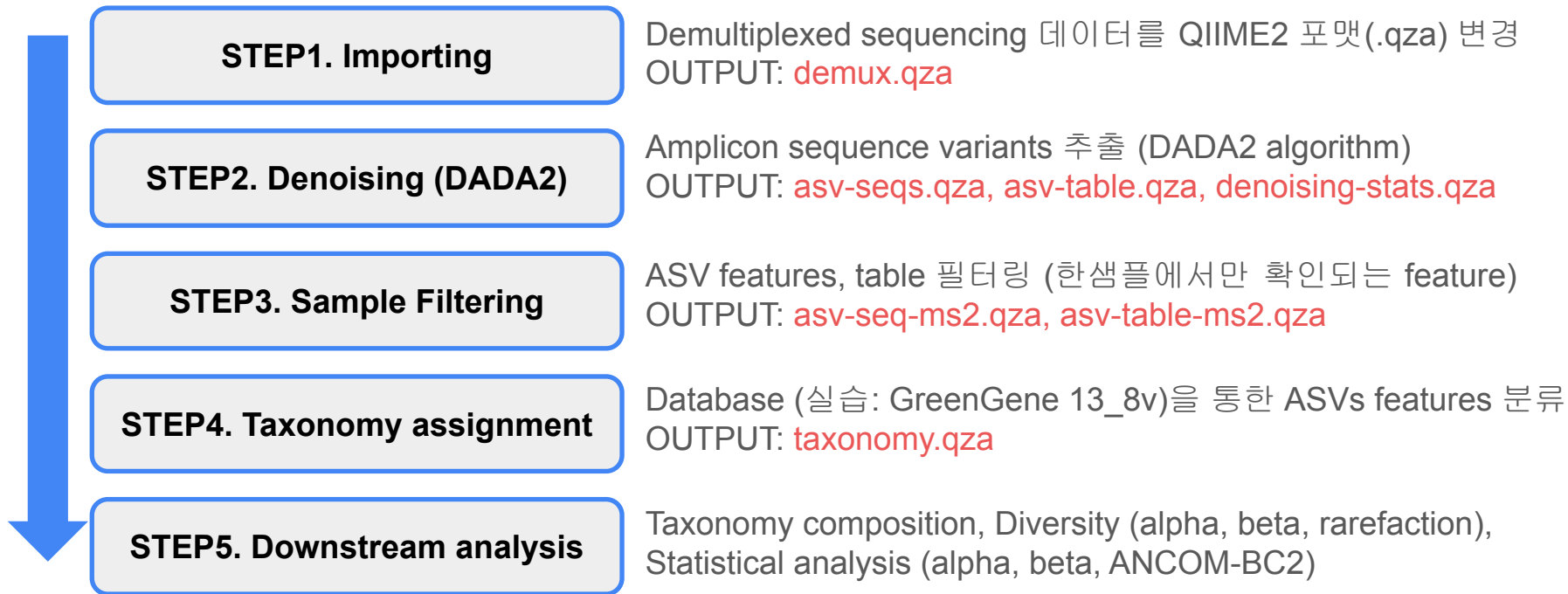
sample-metadata.tsv

sample-id	Bucket	SampleNumber	SampleType	FLWC Detail	FLWC Age Months	FLWC Location	Sampler Initials	Bulking Material Detail	Composting Time Point	Date	Season	Crate Location	Temp Top Bucket C	Temp Bottom Bucket C	Average Ambient Temp	Wt Bucket Compost Kg	Wt Bucket Kg	Wt Compost Kg	Observati Outside Bucket
445427a7	11	70	Human Excrement Compost				LW		31	2022-06-28		20	32.5	34	33.25	5.8	1.4	4.4	fly tape re
e23988ea	12	65	Human Excrement Compost				LW		29	2022-06-28		15	33.5	35	34.25	5.3	1.35	3.95	
5f6857b0	11	69	Human Excrement Compost Pre-Roll				LW		31	2022-06-28		20	31.5	31	31.25	5.8	1.4	4.4	fly tape re
5e994dd4	12	64	Human Excrement Compost Pre-Roll				LW		29	2022-06-28		15	33	33	33	5.3	1.35	3.95	
4276029	9	68	Human Excrement Compost				LW		31	2022-06-28		12	29	30	29.5	9.1	1.35	7.75	fly tape re
77a2bad0	10	70	Human Excrement Compost Pre-Roll				LW		31	2022-06-28		11	30.5	31	30.75	5.6	1.3	4.3	fly tape re
3494909	7	71	Human Excrement Compost Pre-Roll				LW		31	2022-06-28		13	33	34	33.5	5.2	1.35	3.85	

(예시: 일부 열)

- 퇴비화 상태: Compost Temp C (온도), Compost Moisture Percent (수분함량), Compost pH
- 물리적 특성: Sample Weight g (무게), Observed Sample Smell (냄새), Observed Sample Color (색상)
- 기타 환경 데이터: Average Ambient Temp (외기 온습도), Bulking Material Detail (첨가제 종류)

QIIME2 실습



STEP1. QIIME2 Importing

💡 본 실습에서는 시간관계상 Importing 된 **demux.qza** 파일을 활용하겠습니다.

<fq-manifest.tsv>

sample-id absolute-filepath

sample-1 /scratch/microbiome/sample1_R1.fastq.gz

sample-2 /scratch/microbiome/sample2_R1.fastq.gz

```
# Single-End Importing
```

```
qiime tools import \
```

```
--type 'SampleData[SequencesWithQuality]' \
```

```
--input-path fq-manifest.tsv \
```

```
--output-path demux.qza \
```

```
--input-format SingleEndFastqManifestPhred33V2
```

STEP1. QIIME2 Importing

💡 본 실습에서는 시간관계상 Importing 된 **demux.qza** 파일을 활용하겠습니다.

< **fq-manifest.tsv** >

sample-id forward-absolute-filepath reverse-absolute-filepath

sample-1 /scratch/microbiome/sample1_R1.fastq.gz /scratch/microbiome/sample1_R2.fastq.gz

sample-2 /scratch/microbiome/sample2_R1.fastq.gz /scratch/microbiome/sample2_R2.fastq.gz

```
# Paired-End Importing
```

```
qiime tools import \
```

```
--type 'SampleData[PairedEndSequencesWithQuality]' \
```

```
--input-path fq-manifest.tsv \
```

```
--output-path demux.qza \
```

```
--input-format PairedEndFastqManifestPhred33V2
```

QIIME2 실습

STEP1. QIIME2 Importing

💡 본 실습에서는 시간관계상 Importing 된 **demux.qza** 파일을 활용하겠습니다.

```
# manifest.txt 파일 만들기
```

```
bash make_docker_manifest.sh
```

```
# QIIME2 포맷으로 변경
```

```
qiime-docker tools import \
```

```
--type 'SampleData[PairedEndSequencesWithQuality]' \
```

```
--input-path manifest.tsv \
```

```
--output-path demux.qza \
```

```
--input-format PairedEndFastqManifestPhred33V2
```

QIIME2 실습

STEP1. QIIME2 Importing

```
# Summarize Demultiplexed Sequences
qiime-docker demux summarize \
--i-data demux.qza \
--o-visualization output/demux.qzv
```

demux.qzv | [view](#)

QIIME2 실습

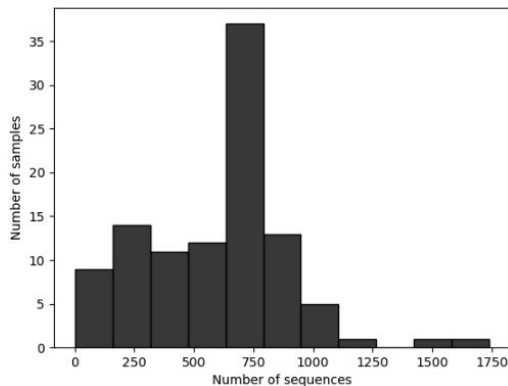
STEP1. QIIME2 Importing

demux.qzv | [view](#)

Demultiplexed sequence counts summary

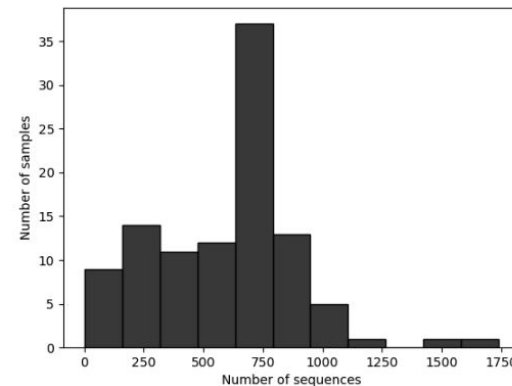
	forward reads	reverse reads
Minimum	0	0
Median	659.5	659.5
Mean	599.788462	599.788462
Maximum	1741	1741
Total	62378	62378

Forward Reads Frequency Histogram



[Download as PDF](#)

Reverse Reads Frequency Histogram



[Download as PDF](#)

QIIME2 실습

STEP1. QIIME2 Importing

demux.qzv | [view](#)

Per-sample sequence counts

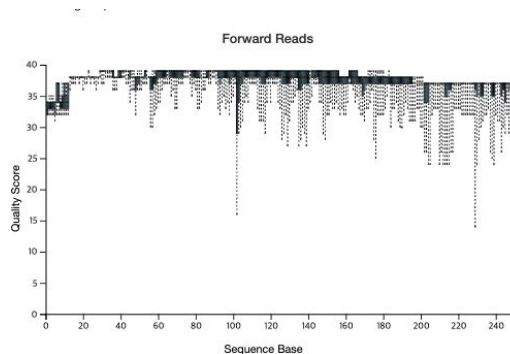
Total Samples: 104 (forward) 104 (reverse)

	forward sequence count	reverse sequence count
sample ID		
016287d9	413	413
027ff912	654	654
05751894	658	658
059c4922	914	914
073e29ef	286	286
0948e1f5	903	903
09fe1	726	726
0e8ff749	567	567
0fc4a04a	524	524
1322edf8	984	984
13ad5b3f	403	403
14aeaa5c	767	767
15cbc	850	850
188af463	320	320
1f51a	721	721
1f98752c	933	933
1fd8f153	565	565

QIIME2 실습

STEP1. QIIME2 Importing

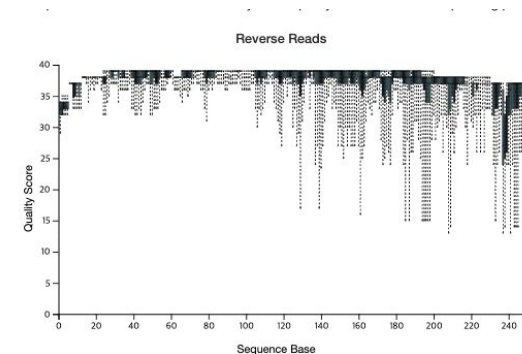
demux.qzv | [view](#)



The plot at position 102 was generated using a random sampling of 10000 out of 62378 sequences without replacement. The minimum sequence length identified during subsampling was 251 bases. Outlier quality scores are not shown in box plots for clarity.

Parametric seven-number summary for position 102		
Box plot feature	Percentile	Quality score
(Not shown in box plot)	2nd	14
Lower Whisker	9th	16
Bottom of Box	25th	29
Middle of Box	50th (Median)	37
Top of Box	75th	39
Upper Whisker	91st	39
(Not shown in box plot)	98th	39

[Download forward parametric seven-number summaries as TSV](#)



The plot at position 161 was generated using a random sampling of 10000 out of 62378 sequences without replacement. The minimum sequence length identified during subsampling was 251 bases. Outlier quality scores are not shown in box plots for clarity.

Parametric seven-number summary for position 161		
Box plot feature	Percentile	Quality score
(Not shown in box plot)	2nd	13
Lower Whisker	9th	16
Bottom of Box	25th	36
Middle of Box	50th (Median)	38
Top of Box	75th	39
Upper Whisker	91st	39
(Not shown in box plot)	98th	39

[Download reverse parametric seven-number summaries as TSV](#)

Demultiplexed sequence length summary

Forward Reads

Total Sequences Sampled	10000.0
2%	251 nts
9%	251 nts
25%	251 nts
50% (Median)	251 nts
75%	251 nts
91%	251 nts
98%	251 nts

Reverse Reads

Total Sequences Sampled	10000.0
2%	251 nts
9%	251 nts
25%	251 nts
50% (Median)	251 nts
75%	251 nts
91%	251 nts
98%	251 nts

STEP2. DADA2 Denoising

DADA2 Denoising

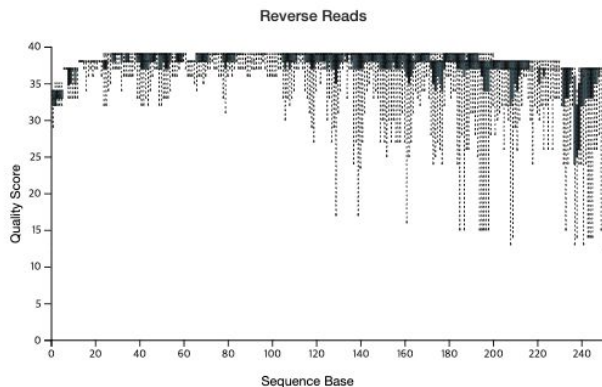
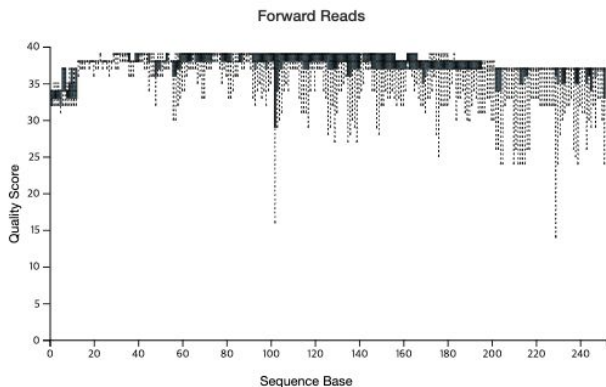
```
qiime-docker dada2 denoise-paired \  
--i-demultiplexed-seqs input/demux.qza \  
--p-trim-left-f 0 \  
--p-trunc-len-f 250 \  
--p-trim-left-r 0 \  
--p-trunc-len-r 250 \  
--p-n-threads 30 \ #분석할 CPU threads 수  
--o-representative-sequences asv-seqs.qza \  
--o-table asv-table.qza \  
--o-denoising-stats denoising-stats.qza \  
--o-base-transition-stats base-transition-stats.qza
```

QIIME2 실습

STEP2. DADA2 Denoising

Q. **trunc-len-f, trunc-len-r** 가 250으로 지정된 이유?

□ 초기 서열의 품질이 좋기 때문에 **250bp** 을 모두 활용하여 진행하기로 함



STEP2. DADA2 Denoising: Analysis Stats

```
# DADA2 Denoising Stats
qiime-docker metadata tabulate \
--m-input-file denoising-stats.qza \
--o-visualization denoising-stats.qzv
```

denoising-stats.qzv | [view](#)

STEP2. DADA2 Denoising: Analysis Stats

denoising-stats.qzv | [view](#)

sample-id #q2-types	input numeric	filtered numeric	percentage of input passed filter numeric	denoised numeric	merged numeric	percentage of input merged numeric	non-chimeric numeric	percentage of input non-chimeric numeric
016287d9	413	359	86.92	303	293	70.94	293	70.94
027f1912	654	596	91.13	521	458	70.03	458	70.03
05751894	658	569	86.47	521	514	78.12	514	78.12
059c4922	914	839	91.79	778	729	79.76	729	79.76
073e29ef	286	267	93.36	120	88	30.77	88	30.77
0948e1f5	903	827	91.58	749	703	77.85	703	77.85
09fe1	726	659	90.77	597	533	73.42	533	73.42
0e8ff749	567	520	91.71	454	409	72.13	409	72.13
0fc4a04a	524	470	89.69	437	407	77.67	407	77.67
1322edf8	984	854	86.79	790	726	73.78	726	73.78
13ad5b3f	403	360	89.33	282	257	63.77	257	63.77
14aeaa5c	767	705	91.92	602	581	75.75	581	75.75
15cbc	850	748	88	354	264	31.06	264	31.06
188af463	320	285	89.06	163	115	35.94	115	35.94
1f51a	721	672	93.2	577	545	75.59	466	64.63
1f98752c	933	851	91.21	742	682	73.1	682	73.1
1fd8f153	565	521	92.21	438	393	69.56	393	69.56
21f2e6d0	669	547	81.76	455	423	63.23	423	63.23
239ad325	656	601	91.62	468	425	64.79	425	64.79
27c41e29	187	160	85.56	119	117	62.57	117	62.57
29a5795e	1034	943	91.2	881	826	79.88	826	79.88

STEP2. DADA2 Denoising : ASVs table

```
# ASVs table Summary
```

```
qiime-docker feature-table summarize \  
--i-table asv-table.qza \  
--m-metadata-file input/sample-metadata.tsv \  
--o-summary asv-table.qzv \  
--o-sample-frequencies sample-frequencies.qza \  
--o-feature-frequencies asv-frequencies.qza
```

[asv-table.qzv](#) | [view](#)

STEP2. DADA2 Denoising : ASVs table

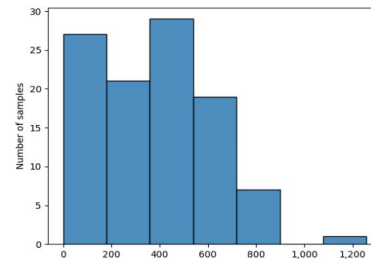
[asv-table.qzv](#) | [view](#)

Table summary

Summary Statistic	Value
Number of samples	104
Number of unique features	1,069
Total observations (total frequency)	39,949

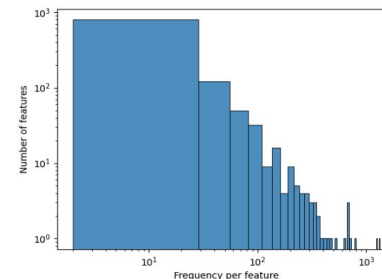
Frequency per sample

	Frequency
Minimum frequency	0
1st quartile	149.8
Median frequency	408
3rd quartile	554.2
Maximum frequency	1,258
Mean frequency	384.1



Frequency per feature

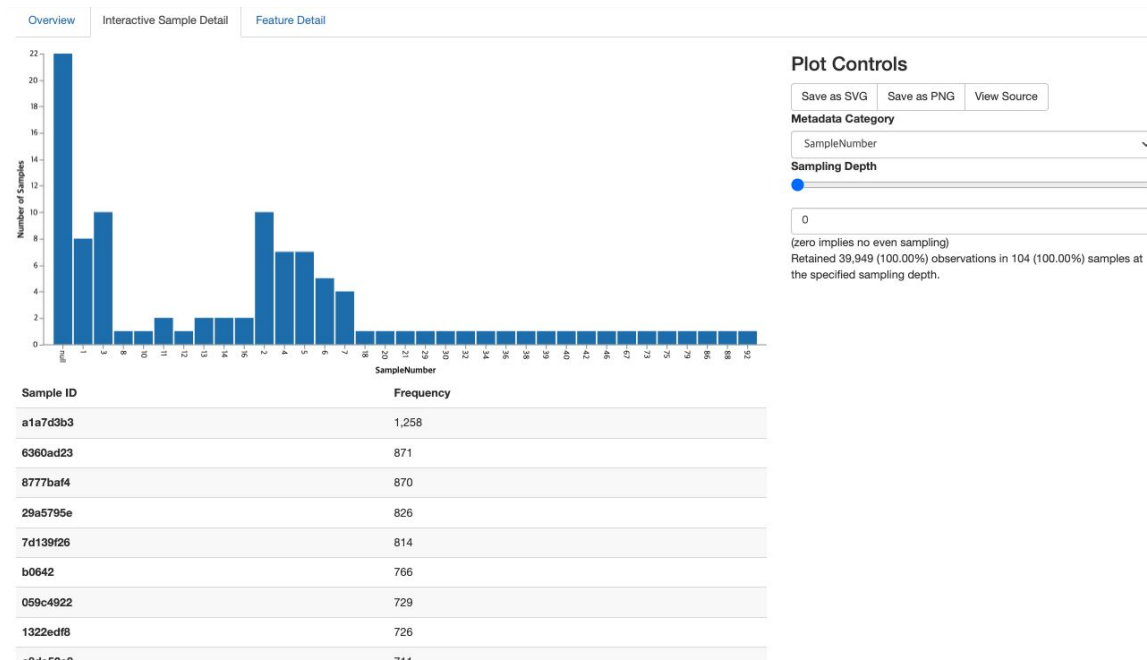
	Frequency
Minimum frequency	2
1st quartile	6
Median frequency	12
3rd quartile	29
Maximum frequency	1,334
Mean frequency	37.4



QIIME2 실습

STEP2. DADA2 Denoising : ASVs table

[asv-table.qzv](#) | [view](#)



STEP2. DADA2 Denoising: ASVs table

asv-table.qzv | [view](#)

Overview	Interactive Sample Detail	Feature Detail		
			Frequency	# of Samples Observed In
c6c3ab4e828fb40d6e05967b7aac9338			1,334	23
430d480477d450614cc88f3853c6ae7f			1,236	27
f5f5e0da89730462abaf6301a9557193			798	19
5be91d78ab696679ced932f4a1e106f8			699	14
99deb3c5ecb022ec05609ebd1112a557			687	15
263e41ea2c25dc87bd2ff5910cf30d40			672	13
d76d59ec71de0e3b22da0c9cd564d41a			670	18
59777186ad2e0947e97615b5d6225136			630	13
7911816f5e81f650f769aba0d5c708cb			525	2
00a96fbd0ac34bfd245f9c24f8737fd			471	15
cb5e5b58a5874db71638f9d8831b2fc			443	1
e335f74033bc634af43ee6baa84fa247			418	23
9f04d4582c00cc1372a4a536aaa64d65			400	7
0a68ec564e278b0246c920ad7de41880			352	10
d12833fed8673a8d32d1adc368a01538			350	6
99e433a3ce4d5290445f668df2c9147e			339	7
1010ac031fa1a71a79b2004905758a12			332	14
63403aa0515deb0039fbb793b870272			325	12
bad9793c5b98ec73936457e0c28acb7e			308	10

STEP2. DADA2 Denoising: ASVs Sequences

```
# ASVs Sequence Summary
qiime-docker feature-table tabulate-seqs \
--i-data asv-seqs.qza \
--m-metadata-file asv-frequencies.qza \
--o-visualization asv-seqs.qzv
```

asv-seqs.qzv | [view](#)

STEP2. DADA2 Denoising: ASVs sequences

asv-seqs.qzv | [view](#)

Sequence Length Statistics

Download sequence-length statistics as a TSV

Sequence Count	Min Length	Max Length	Mean Length	Range	Standard Deviation
1069	250	324	253.33	74	3.36

Seven-Number Summary of Sequence Lengths

Download seven-number summary as a TSV

Percentile:	2%	9%	25%	50%	75%	91%	98%
Length* (nts):	252	253	253	253	253	253	254

*Values rounded down to nearest whole number.

Sequence Table

To BLAST a sequence against the NCBI nt database, click the sequence and then click the View report button on the resulting page.

Download your sequences as a raw FASTA file

Click on a Column header to sort the table.

Feature ID	Frequency	No. of Samples Observed In	Sequence Length	Sequence ▼
c7710d6f144476667c42e65e2ec71053	13.0	1	252	TACGTAGGGAGCGAGCGTTATCCGGATTCTTGGCGTAAAGCGCGGTAGGCGGCCGTC AAGCGGGGTTTCAAATCCAGGGGCTCAACCTCTGGCCGACGCCGA
7fb08fa9b7b2eef6a3db57d0de769227	21.0	2	252	TACGTAGGGGGCTAGCGTTGTCGGGAATCACTGGGCGTAAAGGGTTCGAGCGGAAATGCAAGTCAGATGTA AAGGCAGTAGCTTAACACTACTGTAAAGCATTGAA
92f1720367db58c08a96e9eb9feb416a	21.0	2	253	TACGAAGGGGGCTAGCTGTGCTGGAATCACTGGGCGTAAAGGGCGCGTAGGCGGCCGATTAAAGTCGGGGTGAAAGCCTGTGGCTCAACACAGAATTGCTCTTCA
ddb36489b462b10d57a194ffb0a15b59	23.0	1	279	TGGTAATTGTGTGTCAGCCGCCGCGTAATCGTAAAGACTAGTGTAGTCATCTTTATTAGGTTTAAAGGGTACCTAGACGGTAAATTAACCTCTAAATGAGTAGC
9bd67bc38e24216c8501e65123bf4dcf	17.0	2	254	GACGAACCGTGCGAAGCTTATTCGGAATCACTGGGCTTAAAGCGCGGTAGGCGGATGGGTGCGTCGGCATCTGAAAGCCCCGGCTCAACCGGGGAAGCGGGGCGG
af91ad4292f18b2961374d9e289da9d3	33.0	1	253	TACGAAGGGTGCAAGCGTTAATCGGAATCACTGGGCGTAAAGCGCGGTAGGCGGTTTGTATAAGTTGGATGTGAAAGCCCCGGGCTCAACCTGGGAATTGCATCCAA
3e7548785f8e792f06c7f63e905e9282	26.0	5	253	TACGAAGGTCCCAAGCGTTGTTCGGAATCACTGGGCGTAAAGAGTCTGTAGGCGGTGAAGTAAGTCAGATGTGAAATCCCGGGGCTCAACCTCGGAACCTGCATCCGA
3fe5d9b594a8cb837c98e20f67bfb2ab	5.0	1	254	TACAGAGGGTGCGAGCGTTGTTCGGAATCACTGGGCGTAAAGCGCGCGAGCGGGACTTCAAGTCGGGAGTGAAAGGCCGGGGCTCAACCCGGAGCTGCTTTGG
e053fbf21bc73ea1948801c1acd5df	33.0	2	253	TACGAAGGGGGCTAGCGTTGTTCGGAATCACTGGGCGTAAAGCGCACGTAGGCGGCTTTGTAAAGTTAGAGGTGAAAGCTCGGGGCTCAACTCCAGAATTGCCTTTAA
df14354ec277348c54dc9dd49eb4da0	106.0	4	253	TACGTAGGTGGCAAGCGTTGTCCGGAATCACTGGGTGTAAAGGGAGGTAGGCGGGATATCAAGTCAGAAGTGAAAATACGGGCTCAACTCGTAACCTGCTTTTGA
bb740c10ad31d5779e628bccffdb5015	7.0	1	253	TACGAAGGGGGCTAGCGTTGTTCGGAATCACTGGGCGTAAAGGGCGGTAGGCGGTTGTCAAGTCAGGCGTGAAGCCCCGGGCTCAACCTGGGAACCGCGCTTGA

STEP3. Filtering features from a feature table

Sequence Length Statistics

Download sequence-length statistics as a TSV

Sequence Count	Min Length	Max Length	Mean Length	Range	Standard Deviation
1069	250	324	253.33	74	3.36

Seven-Number Summary of Sequence Lengths

Download seven-number summary as a TSV

Percentile:	2%	9%	25%	50%	75%	91%	98%
Length* (nts):	252	253	253	253	253	253	254

*Values rounded down to nearest whole number.

Sequence Table

To BLAST a sequence against the NCBI nt database, click the sequence and then click the *View report* button on the resulting page.

Download your sequences as a raw FASTA file

Click on a Column header to sort the table.

Feature ID	Frequency	No. of Samples Observed	Sequence Length	Sequence ▼
c7710d6f144476667c42e65e2ec71053	13.0	1	252	TACGTAGGGAGCGAGCGTTATCCGGATTCTGGGCGTAAAGC
7fb08fa9b7b2eef6a3db57d0de769227	21.0	2	252	TACGTAGGGGGCTAGCGTTGTCGGAATCACTGGGCGTAAAGG
92f1720367db58c68a96cecb9feb416a	21.0	2	253	TACGAAGGGGGCTAGCGTTGTCGGAATCACTGGGCGTAAAGG
ddb36489b462b10d57a194ffb0a15b59	23.0	1	279	TGGTAATTGTGTGTCAGCGCCGCGGTAATACGTAAAGACTA
9bd67bc38e2421c68501e65123bf4dcf	17.0	2	254	GACGAACCGTGCAGCGTTATTCGGAATCACTGGGCTAAAGC
af91ad4292f18b2961374d9e289da9d3	33.0	1	253	TACGAAGGGTGCAGCGTTAATCGGAATCTGGGCGTAAAGC
3e7548785f8e792f06c7f63e905e9282	26.0	5	253	TACGAAGGTCCCAAGCGTTGTCGGAATCACTGGGCGTAAAGA
3fe5d9b594a48cb837c98e20f67bf2ab	5.0	1	254	TACAGAGGGTGCAGCGTTGTCGGAATCACTGGGCGTAAAGC
e053fbbf21bc73ea1948801c1acdd5df	33.0	2	253	TACGAAGGGGGCTAGCGTTGTCGGAATCACTGGGCGTAAAGC
df14354ec277348c54dc9dd49eb4da0	106.0	4	253	TACGTAGGTGGCAAGCGTTGTCGGAATCACTGGGCGTAAAGG
bb740c10ad31d5779e628bccf4db5015	7.0	1	253	TACGAAGGGGGCTAGCGTTGTCGGAATCACTGGGCGTAAAGG

- ASV features, ASV table 에서 단 1개의 샘플에서만 확인되는 희귀서열이 많이 존재함. 이러한 서열을 제거하여 진행될 연산 속도 향상

- 1단계: 최소 2개 이상의 샘플에서 등장하는 서열만 남기도록 ASV table 추출 (**feature-table filter-features**)
- 2단계: 1단계 필터링된 ASV table의 서열만 추출 (**feature-table filter-seqs**)

STEP3. Filtering features from a feature table

```
# ASVs table filtering
```

```
qiime-docker feature-table filter-features \
```

```
--i-table asv-table.qza \
```

```
--p-min-samples 2 \
```

```
--o-filtered-table asv-table-ms2.qza
```

```
# Sequence filtering
```

```
qiime-docker feature-table filter-seqs \
```

```
--i-data asv-seqs.qza \
```

```
--i-table asv-tablems2_viz/feature_frequencies.qza \
```

```
--o-filtered-data asv-seqs-ms2.qza
```

STEP3. Filtering features from a feature table

Q. 필터링된 ASVs table, features (=sequences) 을 확인하는 방법은?

- “STEP2. DADA2 Denoising” 후 생성된 **asv-table.qza, asv-seqs.qza** 을 활용한 Plugin 사용하기
- **feature-table summarize, feature-table tabulate-seqs**

QIIME2 실습

STEP3. Filtering features from a feature table

```
# Filtered ASV-table visualization
```

```
qiime-docker feature-table summarize \  
--i-table asv-table-ms2.qza \  
--m-metadata-file input/sample-metadata.tsv \  
--output-dir asv-table-ms2_viz
```

```
# Filtered ASV-features visualization
```

```
qiime-docker feature-table tabulate-seqs \  
--i-data asv-seqs-ms2.qza \  
--m-metadata-file input/sample-metadata.tsv \  
--output-dir asv-seqs-ms2_viz
```

QIIME2 실습

STEP3. Filtering features from a feature table

Q. How many features did we lose as a result of this filter?

: (필터링 전) 1,069개 □ (후) 335개 (743개 제거, 68.7% 필터링됨)

Q. How many total sequences did we lose?

: 39,949개 □ 29,978개, 9,971개 손실 (25.0% 필터링됨)

Table summary

Summary Statistic	Value
Number of samples	104
Number of unique features	1,069
Total observations (total frequency)	39,949

DADA2 결과: [asv-table.qzv](#) | [view](#)

Table summary

Summary Statistic	Value
Number of samples	99
Number of unique features	335
Total observations (total frequency)	29,978

Filtering 결과: [asv-table-ms2_viz/summary.qzv](#) | [view](#)

STEP4. Taxonomy annotation

Trained 된 DB

```
wget -O 'suboptimal-16S-rRNA-classifier.qza' \  
'https://gut-to-soil-tutorial.readthedocs.io/en/2026.4/data/gut-to-soil/subopti  
mal-16S-rRNA-classifier.qza'
```

suboptimal-16S-rRNA-classifier.qza | **download**

QIIME2 실습

< 데이터베이스를 활용한 Q2-format trained DB 만들기 >

💡 본 실습에서는 시간관계상 trained 된 데이터베이스 활용하겠습니다.

Reference DB 다운로드 - 서열

```
wget -O 'reference-sequences.qza' \  
'https://gut-to-soil-tutorial.readthedocs.io/en/2026.4/data/gut-to-soil/referenc  
e-sequences.qza'
```

Reference DB 다운로드 - 미생물 명명

```
wget -O 'reference-taxonomy.qza' \  
'https://gut-to-soil-tutorial.readthedocs.io/en/2026.4/data/gut-to-soil/referenc  
e-taxonomy.qza'
```

QIIME2 실습

< 데이터베이스를 활용한 Q2-format trained DB 만들기 >

💡 본 실습에서는 시간관계상 trained 된 데이터베이스 활용하겠습니다.

```
# Reference DB Training
```

```
qiime feature-classifier fit-classifier-naive-bayes\  
--i-reference-reads reference-sequences.qza \  
--i-reference-taxonomy reference-taxonomy.qza \  
--o-classifier suboptimal-16S-rRNA-classifier.qza
```

QIIME2 실습

< 데이터베이스를 활용한 Q2-format trained DB 만들기 >

💡 본 실습에서는 시간관계상 trained 된 데이터베이스 활용하겠습니다.

Trained Reference DB 의 결과 확인

```
qiime feature-table tabulate-seqs \  
--i-data reference-sequences.qza \  
--i-taxonomy Greengenes_13_8_85p_OTUs:reference-taxonomy.qza \  
--p-merge-method intersect \  
--o-visualization reference-taxonomy.qzv
```

reference-taxonomy.qzv | [view](#)

QIIME2 실습

< 데이터베이스를 활용한 Q2-format trained DB 만들기 >

 본 실습에서는 시간관계상 trained 된 데이터베이스 활용하겠습니다.

reference-taxonomy.qzv

view

Sequence Length Statistics

Download sequence-length statistics as a TSV

Sequence Count	Min Length	Max Length	Mean Length	Range	Standard Deviation
5088	1258	2353	1430.19	1095	71.0

Seven-Number Summary of Sequence Lengths

Download seven-number summary as a TSV

Percentile:	2%	9%	25%	50%	75%	91%	98%
Length* (nts):	1323	1351	1382	1423	1475	1512	1553

*Values rounded down to nearest whole number.

Sequence Table

To BLAST a sequence against the NCBI nt database, click the sequence and then click the View report button on the resulting page.

Download your sequences as a raw FASTA file

Click on a Column header to sort the table.

Feature ID	Taxon: Greengenes_13.8_85p_OTUs	Sequence Length	Sequence
4451714	k__Bacteria; p__Bacteroidetes; c__Bacteroidia; o__Bacteroidales; f__g__; s__	1400	TTAGAGTTTGATCCTGGCTCAAGATGAACGCTTCGGGAGGCTTAACACATGCAAGTCGAGGGGACGACAAGTATTATATTTGGTGGCGACCGCGGATGGTGAGTAACACGTATGCA
271332	k__Bacteria; p__Actinobacteria; c__OPB41; o__; f__; g__; s__	1383	GACGAGTGGCGAACGGGTGAGTAACACGTGGTAACTGCCCTAAAGACAGGATAACCTTGGGAACAGGGCTAATACCGGATAATGCTGTTTTTCATATGATTCAGACCAATGCTC
240160	k__Bacteria; p__Chloroflexi; c__Ktedonobacteria; o__Thermogemmatissporales; f__Thermogemmatissporaceae; g__; s__	1359	TGCAAGTCGCACGCTCTCTTTTTCTTCGGAAGAGGGAGGGAGTGCCGACGGGTGAGTAACACGTGGTGACCTACCCCAAGTGAGGGATACCGGGCGAAGGCTTGACAAAGCCCGCA
4377099	k__Bacteria; p__Proteobacteria; c__Alphaproteobacteria; o__Rhizobiales; f__Bradyrhizobiaceae; g__; s__	1421	CCTAACACATGCAAGTCGAACGCATCCTTCGGGGTGGTGGCAGCGGTGAGTAACACGTGGGAACGTGCCCATCGTTTCGGGATAACCGAGGAACCTTGGGCTAATACCGGATACGTG

STEP4. Taxonomy annotation

```
# ASVs features Classification
```

```
qiime-docker feature-classifier classify-sklearn \  
--i-classifier input/suboptimal-16S-rRNA-classifier.qza \  
--i-reads asv-seqs-ms2.qza \  
--o-classification taxonomy.qza
```


STEP4. Taxonomy annotation: Summary

Assigned ASVs Visualization

```
qiime-docker feature-table tabulate-seqs \  
--i-data asv-seqs-ms2.qza \  
--i-taxonomy Greengenes_13_8:taxonomy.qza \  
--m-metadata-file asv-frequencies-ms2.qza \  
--o-visualization asv-seqs-ms2.qzv
```

asv-seqs-ms2.qzv | [view](#)

STEP4. Taxonomy annotation: Summary

asv-seqs-ms2.qzv | [view](#)

Sequence Length Statistics

[Download sequence-length statistics as a TSV](#)

Sequence Count	Min Length	Max Length	Mean Length	Range	Standard Deviation
335	251	254	252.98	3	0.26

Seven-Number Summary of Sequence Lengths

[Download seven-number summary as a TSV](#)

Percentile:	2%	9%	25%	50%	75%	91%	98%
Length* (nts):	252	253	253	253	253	253	253

*Values rounded down to nearest whole number.

Sequence Table

To BLAST a sequence against the NCBI nt database, click the sequence and then click the [View report](#) button on the resulting page.

[Download your sequences as a raw FASTA file](#)

Click on a Column header to sort the table.

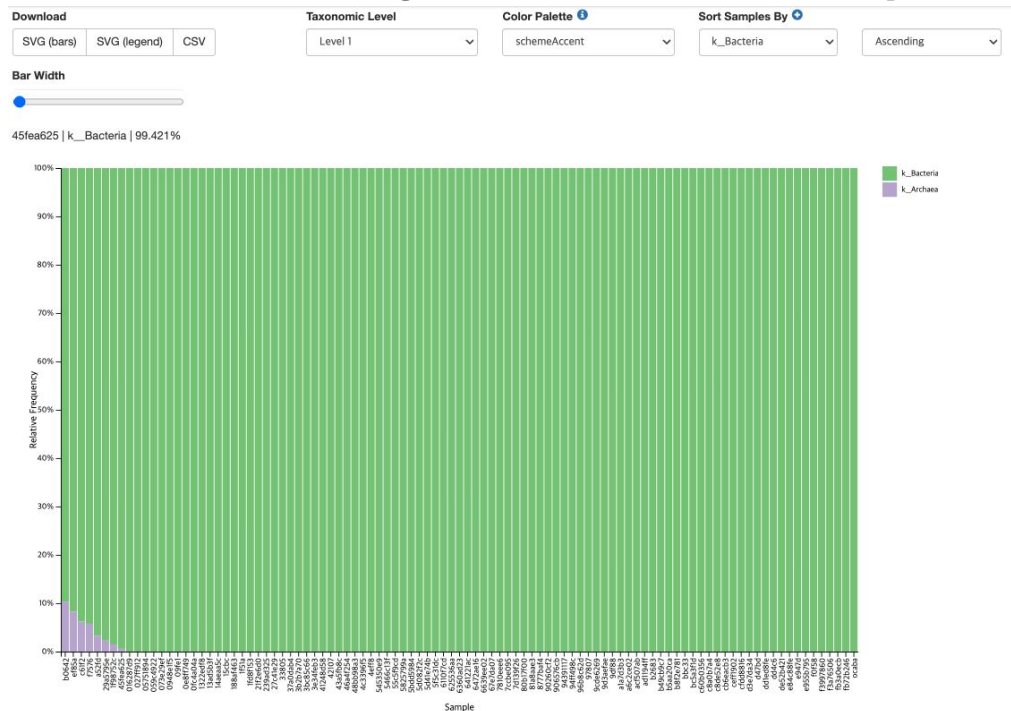
Feature ID	Taxon: Greengenes_13_8	Frequency	No. of Samples Observed	Sequence Length	Sequence
48701661c9172bbfc9dd16f8f43e9519	k__Bacteria; p__Proteobacteria	25.0	2	253	TACGGAGGGAGCTAGCGTTGTTCCGAATTACTGGGCGTAAAGCGTACGTAGCGGTGACTCAAGTCAGAGGTGAAAGCCCGGAGC
7fb08fa9b7b2eef6a3db57d0de769227	k__Bacteria; p__Firmicutes; c__Clostridia; o__Clostridiales	21.0	2	252	TACGTAGGGGGCTAGCGTTGTCGGAATCACTGGGCGTAAAGGGTTCGAGGCGAAATGCAAGTCAGATGTAAAGGCAGTAGC
73f6ea8d82054e97789dedffa06f3de0	k__Bacteria; p__Proteobacteria; c__Gammaproteobacteria	12.0	2	253	TACGGAGGGTGCAAGCGTTAATCGGAATTACTGGGCGTAAAGGGTGCAGTGGTGTAAAGTTATCTGTGAAATCCCCGGGC
078e0fd51a7d5e5b908328591895ba58	k__Bacteria; p__Proteobacteria; c__Alphaproteobacteria; o__RF32; f__; g__; s__	41.0	5	253	TACGAAGGGGGCAAGCGTTGTTCCGAATTACTGGGCGTAAAGGGTGTAGGCGGATTTGTAAGATAGTGGTGAATACCTGAGC
145e42f90511a48e84b834c7e38f678a	k__Bacteria; p__Proteobacteria;	61.0	5	253	TACGAAGGGTGCAAGCGTTACTCGGAATTACTGGGCGTAAAGCGTGCAGTGGTGGTGTGTTAAGTCGTGCGTAAAGCCCTGGGC

STEP5. Downstream data analysis: Taxonomic composition

```
# Taxonomic composition
qiime-docker taxa barplot \
--i-table asv-table-ms2.qza \
--i-taxonomy taxonomy.qza \
--m-metadata-file sample-metadata.tsv \
--o-visualization taxa-bar-plots.qzv
```

[taxa-bar-plots.qzv](#) | **view**

STEP5. Downstream data analysis: Taxonomic composition



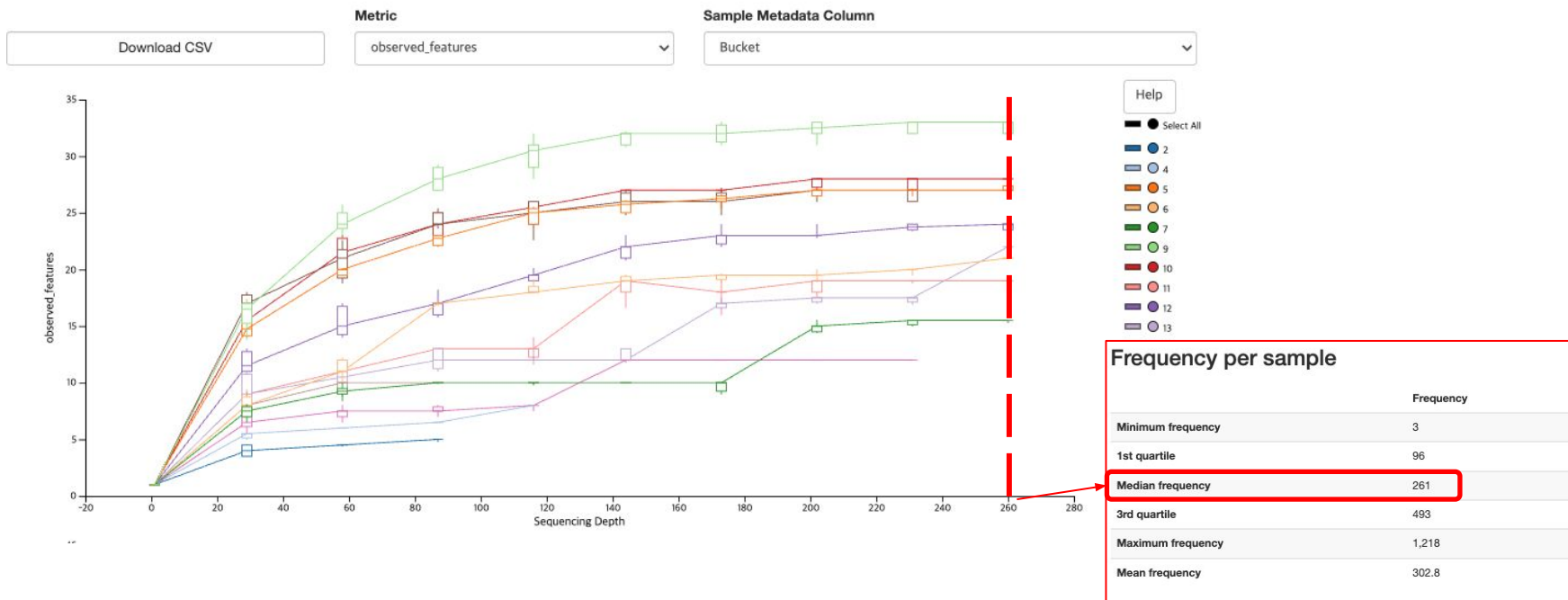
STEP5. Downstream 분석: Rarefaction curve

Normalization 기준치 확인

```
qiime diversity alpha-rarefaction \  
  --i-table asv-table-ms2.qza \  
  --p-max-depth 260 \  
  --m-metadata-file sample-metadata.tsv \  
  --o-visualization alpha-rarefaction.qzv
```

alpha-rarefaction.qzv | [view](#)

STEP5. Downstream 분석: Rarefaction curve



Filtering 결과:

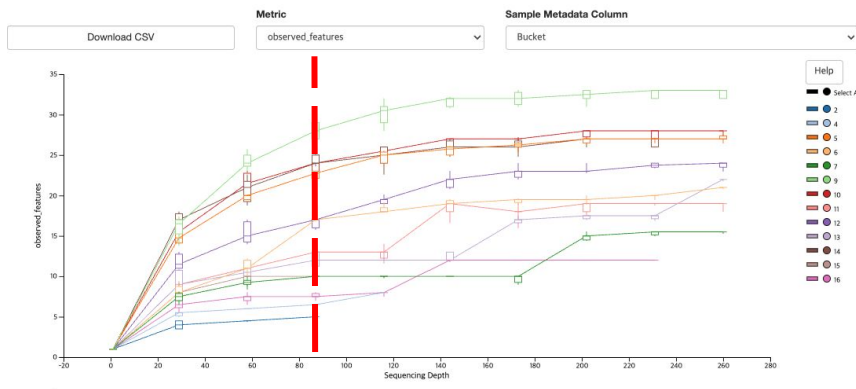
asv-table-ms2.qza

STEP5. Downstream 분석: Diversity

```
# Resampling-based Diversity
qiime boots kmer-diversity \
  --i-table asv-table-ms2.qza \
  --i-sequences asv-seqs-ms2.qza \
  --m-metadata-file sample-metadata.tsv \
  --p-sampling-depth 96 \
  --p-n 10 \
  --p-replacement \
  --p-alpha-average-method median \
  --p-beta-average-method medoid \
  --output-dir boots-kmer-diversity
```

STEP5. Downstream 분석: Diversity

- **q2-boots**: Rarefaction의 무작위 오차를 극복하기 위해 복원 추출(--p-replacement)로 10회 (--p-n 10) 반복 계산 후 중간값(Median/Medoid)을 활용.
- **q2-kmerizer**: 계통수(Phylogenetic tree) 생성 과정을 생략하고 서열을 k-mer 조각으로 분해해 빠른 속도로 계통학적 거리에 준하는 정교한 다양성을 도출.
- **--p-sampling-depth 96**: Frequency per sample의 1st Quartile(96) 지점이자 Rarefaction curve가 안정화되는 지점으로, 전체 샘플의 75%를 보존하면서 최적의 데이터 신뢰성을 확보함.



Frequency per sample

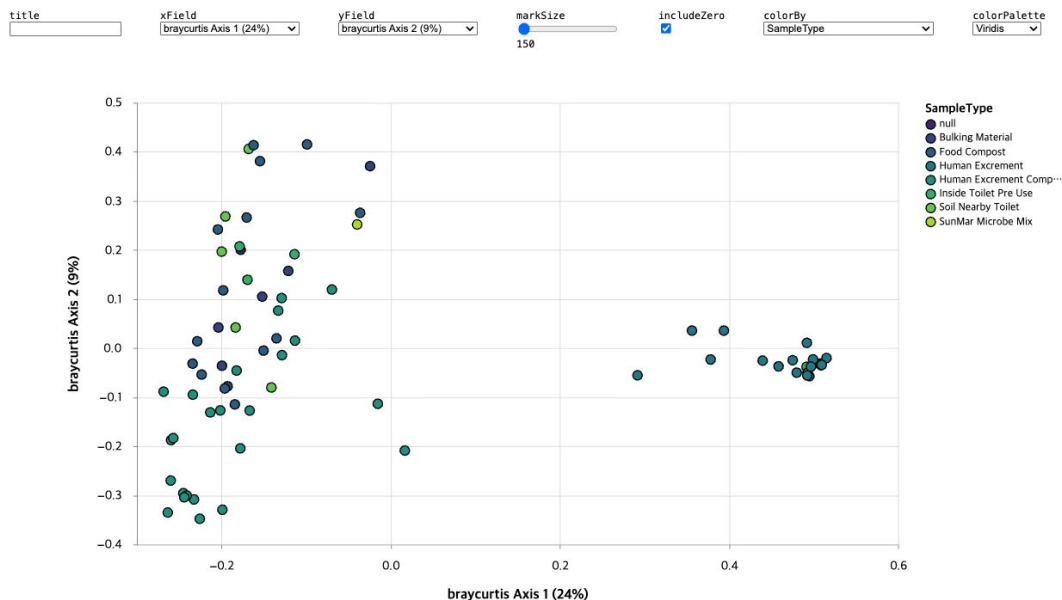
	Frequency
Minimum frequency	3
1st quartile	96
Median frequency	261
3rd quartile	493
Maximum frequency	1,218
Mean frequency	302.8

Filtering 결과:

asv-table-ms2.qza

STEP5. Downstream 분석: Diversity

- [boots-kmer-diversity/scatter_plot.qzv](#) | [view](#)
- [boots-kmer-diversity/resampled_tables/](#)
- [boots-kmer-diversity/kmer_tables/](#)
- [boots-kmer-diversity/alpha_diversities/](#)
- [boots-kmer-diversity/distance_matrices/](#)
- [boots-kmer-diversity/pcoas/](#)



STEP5. Downstream 분석: Diversity

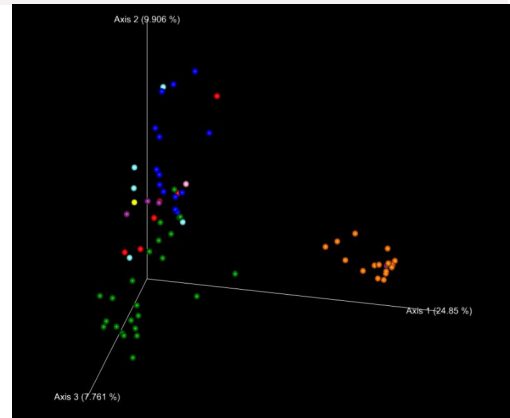
Beta-diversity PCoA plot 시각화하기

qiime-docker emperor plot \

--i-pcoa boots-kmer-diversity/pcoas/braycurtis.qza \

--m-metadata-file input/sample-metadata.tsv \

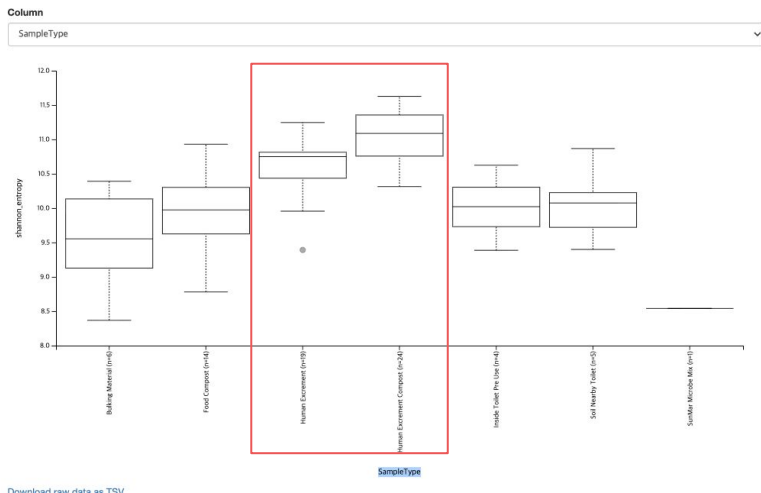
--o-visualization boots-kmer-diversity/ pcoas/braycurtis-emperor.qzv



STEP5. Downstream 분석: Alpha-diversity 그룹간 비교

```
# Alpha 다양성 그룹간 비교 (qiime diversity alpha-group-significance)
qiime-docker diversity alpha-group-significance \
  --i-alpha-diversity boots-kmer-diversity/alpha_diversities/shannon.qza \
  --m-metadata-file input/sample-metadata.tsv \
  --o-visualization boots-kmer-diversity/alpha_diversities/shannon-group-significance.qzv
```

STEP5. Downstream 분석: Alpha-diversity 그룹간 비교



Kruskal-Wallis (all groups)

	Result
H	41.80400506078229
p-value	2.0101001789067548e-7

Kruskal-Wallis (pairwise)

[Download CSV](#)

		H	p-value	q-value
Group 1	Group 2			
Bulking Material (n=6)	Food Compost (n=14)	0.979592	0.322300	0.423018
	Human Excrement (n=19)	9.327935	0.002257	0.009479
	Human Excrement Compost (n=24)	13.172043	0.000284	0.002984
	Inside Toilet Pre Use (n=4)	0.727273	0.393769	0.459397
	Soil Nearby Toilet (n=5)	0.833333	0.361310	0.446325
Food Compost (n=14)	SunMar Microbe Mix (n=1)	1.000000	0.317311	0.423018
	Human Excrement (n=19)	10.987616	0.000917	0.006421
	Human Excrement Compost (n=24)	21.157509	0.000004	0.000089
	Inside Toilet Pre Use (n=4)	0.011278	0.915425	0.915425
	Soil Nearby Toilet (n=5)	0.077143	0.781207	0.846821
Human Excrement (n=19)	SunMar Microbe Mix (n=1)	2.625000	0.105193	0.184087
	Human Excrement Compost (n=24)	8.469498	0.003612	0.012640
	Inside Toilet Pre Use (n=4)	5.157895	0.023141	0.060745
	Soil Nearby Toilet (n=5)	3.548211	0.059610	0.139089
	SunMar Microbe Mix (n=1)	2.714286	0.099453	0.184087
Human Excrement Compost (n=24)	Inside Toilet Pre Use (n=4)	7.969828	0.004756	0.014269
	Soil Nearby Toilet (n=5)	9.363333	0.002214	0.009479
	SunMar Microbe Mix (n=1)	2.769231	0.096092	0.184087
Inside Toilet Pre Use (n=4)	Soil Nearby Toilet (n=5)	0.060000	0.806496	0.846821
	SunMar Microbe Mix (n=1)	2.000000	0.157299	0.235949
Soil Nearby Toilet (n=5)	SunMar Microbe Mix (n=1)	2.142857	0.143235	0.231379

인분(HE)이 퇴비화(HEC) 과정을 거치면서 환경 미생물의 유입과 유기물 분해 작용에 의해 종 다양성(Shannon)이 최고 수치로 증가하여, 미생물 생태계가 훨씬 풍부하고 복잡하게 발전했음을 보여줌. ($q=0.013$)

STEP5. Downstream 분석: Beta-diversity 그룹간 비교

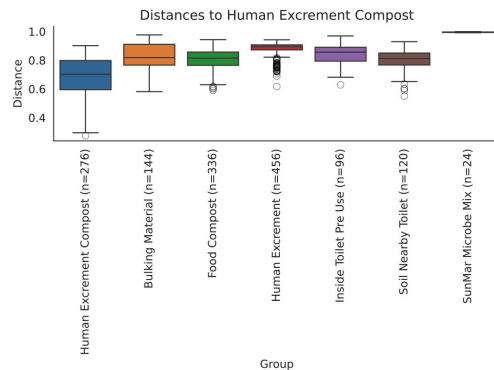
Beta 다양성 그룹간 비교

```
qiime-docker diversity beta-group-significance \  
--i-distance-matrix boots-kmer-diversity/distance_matrices/braycurtis.qza \  
--m-metadata-file input/sample-metadata.tsv \  
--m-metadata-column SampleType \  
--p-pairwise \  
--o-visualization  
boots-kmer-diversity/distance_matrices/bray-curtis-sample-type-significance.qzv
```

STEP5. Downstream 분석: Beta-diversity 그룹간 비교

Overview

PERMANOVA results	
method name	PERMANOVA
test statistic name	pseudo-F
sample size	73
number of groups	7
test statistic	6.432746
p-value	0.001
number of permutations	999



Pairwise permanova results

[Download CSV](#)

		Sample size	Permutations	pseudo-F	p-value	q-value
Group 1	Group 2					
Bulking Material	Food Compost	20	999	1.395924	0.109	0.152600
	Human Excrement	25	999	11.381932	0.001	0.003500
	Human Excrement Compost	30	999	2.281657	0.006	0.015750
	Inside Toilet Pre Use	10	999	0.726198	0.830	0.832000
	Soil Nearby Toilet	11	999	0.918546	0.577	0.637737
	SunMar Microbe Mix	7	999	0.987417	0.832	0.832000
Food Compost	Human Excrement	33	999	21.193257	0.001	0.003500
	Human Excrement Compost	38	999	6.020122	0.001	0.003500
	Inside Toilet Pre Use	18	999	2.244296	0.008	0.016800
	Soil Nearby Toilet	19	999	1.822180	0.044	0.066000
	SunMar Microbe Mix	15	999	2.444294	0.025	0.047727
Human Excrement	Human Excrement Compost	43	999	26.352985	0.001	0.003500
	Inside Toilet Pre Use	23	999	6.635675	0.003	0.009000
	Soil Nearby Toilet	24	999	13.814755	0.001	0.003500
	SunMar Microbe Mix	20	999	7.783879	0.034	0.054923
Human Excrement Compost	Inside Toilet Pre Use	28	999	2.535921	0.008	0.016800
	Soil Nearby Toilet	29	999	2.959739	0.001	0.003500
	SunMar Microbe Mix	25	999	3.102843	0.034	0.054923
Inside Toilet Pre Use	Soil Nearby Toilet	9	999	0.996188	0.468	0.546000
	SunMar Microbe Mix	5	999	1.020448	0.426	0.526235
Soil Nearby Toilet	SunMar Microbe Mix	6	999	1.880835	0.179	0.234937

STEP5. Downstream 분석: Beta-diversity 그룹간 비교

- Pairwise PERMANOVA 분석 결과, 인분(HE)과 인분 퇴비(HEC) 간의 군집 차이는 ***pseudo-F = 26.35, q = 0.0035***
- 전체 비교 그룹 중 가장 높은 분리도(pseudo-F)와 명확한 통계적 유의성을 보였음.
- 이는 퇴비화 과정을 통해 장내 유래 미생물군이 완전히 다른 환경형 미생물 군집으로 성공적으로 재편되었음

STEP5. Downstream 분석: Differential abundance (ANCOM-BC2)

(1) 주요 샘플 그룹 선별

```
qiime feature-table filter-samples \  
  --i-table asv-table-ms2.qza \  
  --m-metadata-file sample-metadata.tsv \  
  --p-where '[SampleType] IN ("Human Excrement Compost", "Human Excrement", "Food Compost")' --o-filtered-table asv-table-ms2-dominant-sample-types.qza
```

(2) Genus level 에 해당되는 ASV table 추출

```
qiime-docker taxa collapse \  
  --i-table asv-table-ms2-dominant-sample-types.qza \  
  --i-taxonomy taxonomy.qza \  
  --p-level 6 \  
  --o-collapsed-table genus-table-ms2-dominant-sample-types.qza
```


STEP5. Downstream 분석: Differential abundance (ANCOM-BC2)

(3) ANCOM-BC2 검정 실행

```
qiime-docker composition ancombc2 \  
  --i-table genus-table-ms2-dominant-sample-types.qza \  
  --m-metadata-file sample-metadata.tsv \  
  --p-fixed-effects-formula SampleType \  
  --p-reference-levels 'SampleType::Human Excrement Compost' \  
  --o-ancombc2-output genus-ancombc2-results.qza
```

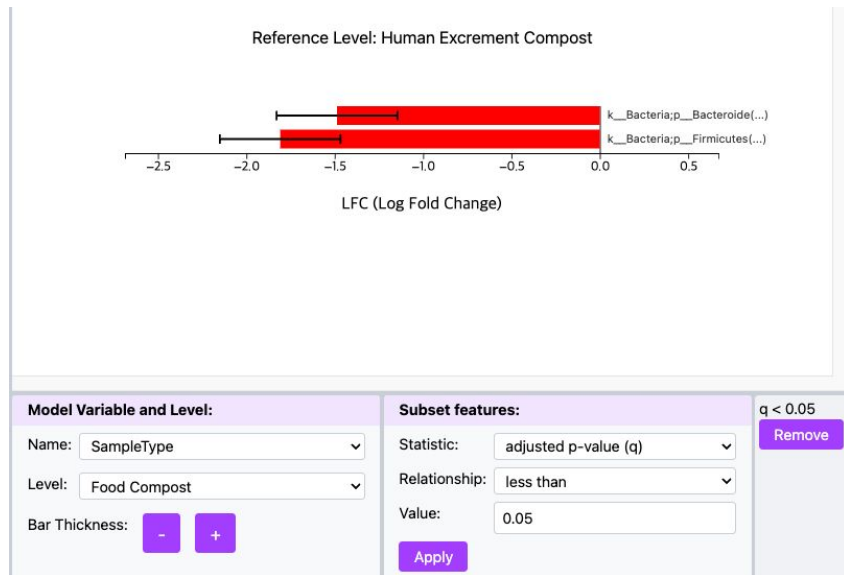
(4) Visualization

```
qiime-docker composition ancombc2-visualizer \  
  --i-data genus-ancombc2-results.qza \  
  --o-visualization genus-ancombc2-barplot.qzv
```

[genus-ancombc2-barplot.qzv](#) | [view](#)

STEP5. Downstream 분석: Differential abundance (ANCOM-BC2 응용)

[genus-ancombc2-barplot.qzv](#) | [view](#)



이전결과와 다른 점은?

- **Collapse** 미적용 시, 전체 **ASV** 개수에 대하여 가설 검정이 진행되며 **FDR** 보정 기준이 엄격해짐.
- 특정 **Taxonomic rank** 에서 확인하고 싶을때에는 **Collapse** 후 **ANCOM-BC2** 진행 필요

HNB Portal

휴엔바이옴 HNB 포탈 메타지놈 분석 서비스 v1.0 사용자 가이드 |
HNB Portal Metagenome Analysis Service v1.0 User Guide



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